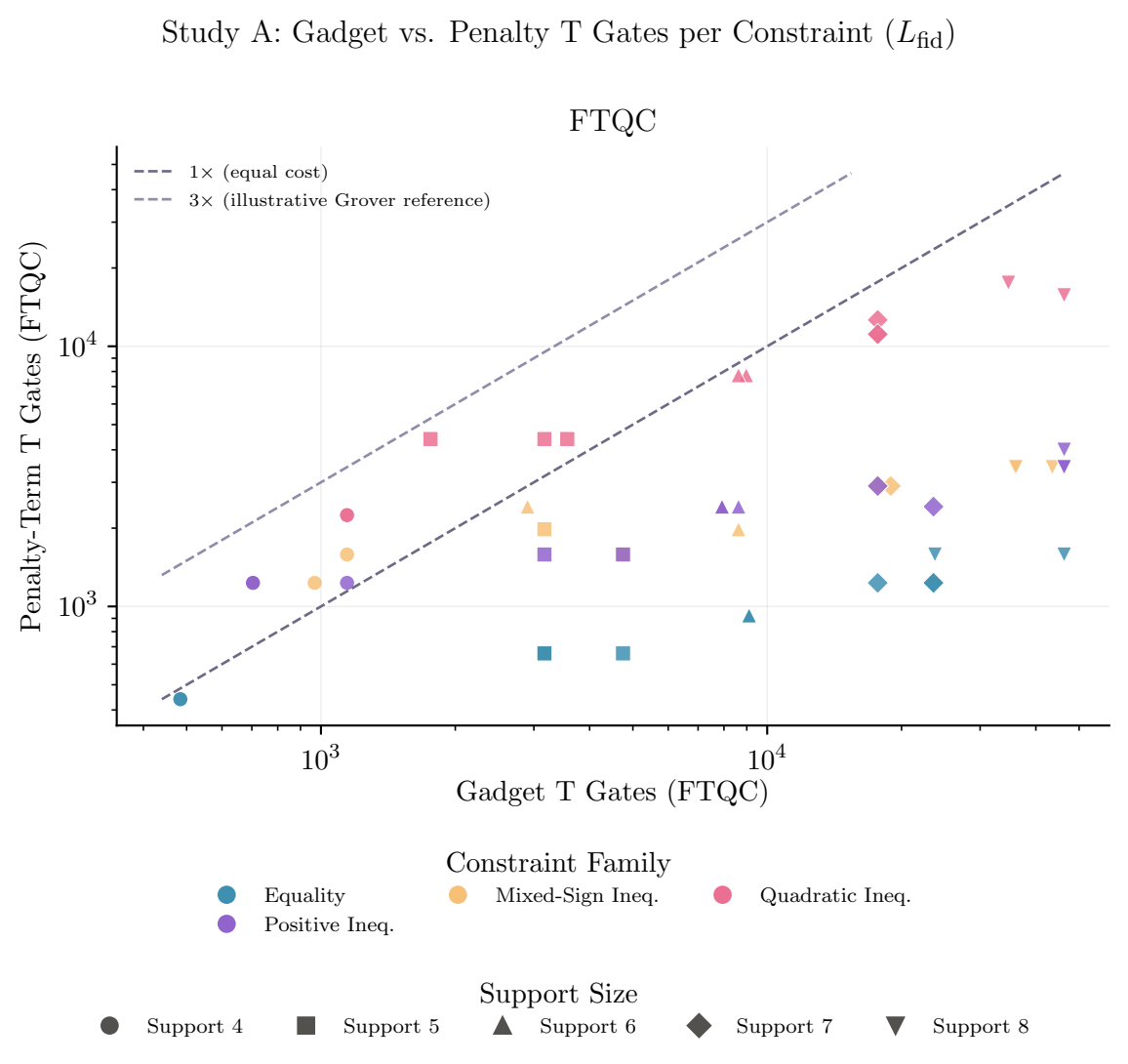

Study A: Gadget vs. Penalty T Gates per Constraint (L_{fid})
FTQC
1× (equal cost)
3× (illustrative Grover reference)
Penalty-Term T Gates (FTQC)
10^4
10^3
10^3
10^4
Gadget T Gates (FTQC)
Constraint Family
Equality
Mixed-Sign Ineq.
Quadratic Ineq.
Positive Ineq.
Support Size
Support 4
Support 5
Support 6
Support 7
Support 8